* GROW IT YOURSELF *

diy guide *

SCULPTING WITH



MYCELIUM



grow from mushrooms

exploring and growing mushrooms

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introduction

Mycelium is the root structure of mushrooms. The mushroom part you actually see is just a fleeting fruiting body of the organism; the mycelium itself is underground.



Sebastian Cox

Many things can be created out of this biodegradable living material.

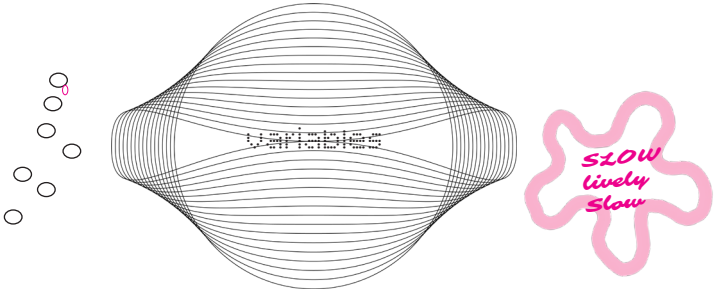


Mycelium Millenium

Mycelium has anti-bacterial, heat resistant and flame retardant properties. It can be used for packaging, insulation, and making of all kinds! This guide will explore how you can get started making your own forms with mushrooms, whether you want to make packaging for your small business, art, or objects for your house.

things to think about

When working with a living thing there is a lot that can go astray, a lot of potential surprises and a lot to think about! You are ultimately collaborating with a creature motivated by its own needs and navigating useseen fungal worlds. Factors such as humidity, airflow and spores/bacteria floating around your environment will all play a role in the growth of your mycelium.



In contrast to industrial materials that are fast, easy and reliable, mycelium is slow, lively and dynamic.
What do you expect to learn from adjusting your expec experience of working with a living creature?

supplies

☐ latex gloves



☐ stove or instant pot



☐ flour



☐ rubbing alcohol



☐ a mushroom
☐ air port containers
☐ substrate

*

There is no one way to grow mycelium. If you have a pre-innoculated substrate from Grow Bio or a mushroom grower, skip to the mold section --->.

* **a fresh mushroom or spore innoculate:** You can start with any foraged or purchased non-toxic mushroom you want! Different mushrooms have different strenghts, resilence and growth preferences. Oyster mushrooms are great for mycelium cultivation and this guide will focus on them.

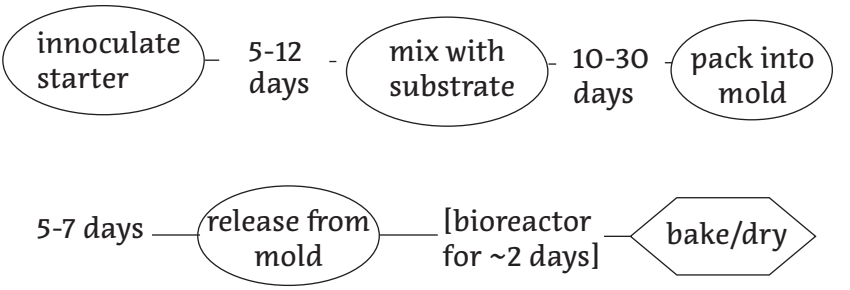
* **air port containers:** you will want to make or buy containers that filter out pollutants. These can either be jars or bags. To make bags:

* **substrate:** exploring different substrates is a great way to experiement with materials. Substrate is the food/structure the mycelium grows in; mycelium grows in between the fibers and turns the whole thing into a solid structure.

Choosing a substrate -- Most substrates are loose organic materials like wood chips, straw, hemp husks, shredded cardboard, or whole oat grain. The look and feel of your substrate will effect your final form. You can run some experiments to discover different substrates in small jars and see what your mycelium prefers. Substrate must be pasturized (using a pressure cooker) or sterilized (steamed on the stove for two hours and allowed to cool) to reduce contamination and give your mycelium its best shot at life.



growth stages overview



INNOCULATE STARTER: There are several ways to do this. You can take a spore print of a mushroom and grow the spores in pertri dishes, innoculate a liquid solution, or grow on a piece of cardboard. The easiest thing to do is to place a slice of mushroom on sterilized carboard in a tupperware.

1. first, boil a strip of cardboard from a toilet paper or paper towel roll for 20 minutes to sterilize it.

2. spray the inside of a tupperware with rubbing alcohol and then add the cardboard, taking care to lift the lid as briefly as possible to prevent airborne contamination.

3. pass the end of a knife through a flame. Tear open your mushroom and take several strips of flesh from its inside. Place these on the cardboard.



4. Cover with the lid and wait. Within 10 days, you should see the white mycelium hyphae growing from your sample.

MIX WITH SUBSTRATE: Once you have chosen a substrate, you'll need to sterilize or pasturize it. Pasturizing substrate takes longer and is more effective. For this guide, we'll just sterilize it. You can either steam it for 10 minutes in a pressure cooker or instant pot or boil it for two hours on the stove.

1. If you steam it in an instant pot, first soak it for a few minutes then drain, then add back 2 cups of water, which is the minimum amount the instant pot needs to create steam.

Once your substrate is sterilized, drain the excess water. Then, add it to the air port container of your choice (I used a store-bought grow bag). Finally, pass a knife through a flame and use it to transfer your mycelium starter from the tupperware into the substrate.
Seal the container and put your innoculated substrate into a closet for the next week or two, checking for white mycelium growth. If your substrate begins to mold a color that is not white, you need to throw it away and start over.





PACK INTO MOLD:
While your mycelium is growing, you have plenty of time to construct a mold for its final form. Flip to page x to see a guide on how to choose a mold. Once your substrate is fairly consistently white with mycelium but before it solidifies into the shape of the bag, the mycelium is ready to pack into your mold.

1. Spray the mold and your area with rubbing alcohol and put on latex gloves.

2. If your mold is made of cardboard or some other material the mycelium may begin to eat and bind to, line it with plastic wrap.

3. remove the mycelium from the bag, breaking up the clumps of substrate that have formed and pack into the mold.

4. the white mycelium may dissapear as the links are broken, turning your substrate back into its original color. Don't worry about that, it's still innoculated and will grow back!

5. Place the entire mold back into a air port bag if it fits, otherwise use a large lidded container. You can create your own bioreactor with a cardboard box and a plastic wrap cover - be sure to use clean cardboard and spray everything down.

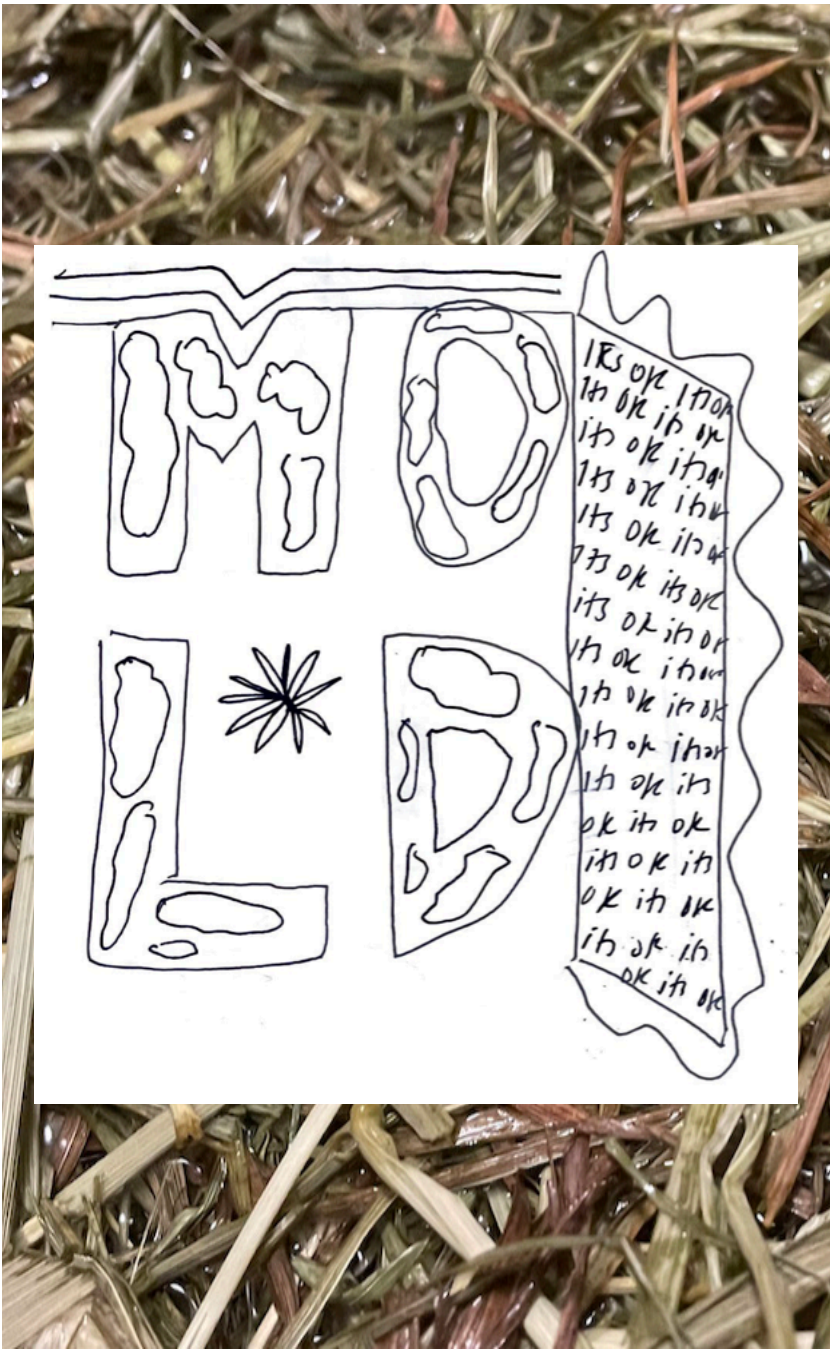
6. After 3-5 days, remove the form from the mold. If you want it to be fluffier, you can sprinkle flour and spray water on the form and place it in the bioreactor for another few days.

growth stages log

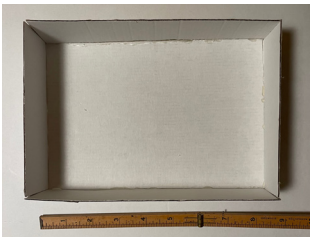
mycelium love letter



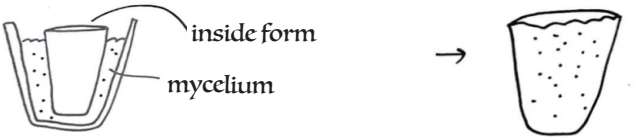
my straw substrate molded on day 12.


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choosing a mold



You can make a custom mold out of cardboard and hot glue or using 3D objects in your house. Some amount of need pressue is needed to form the shape, and mycelium that is too thick for air/water flow (more than a few inches) might not stay together. To make a hollow 3D shape, choose two objects that fit inside eachtoher and fill the space in between with mycelium.







experiments in growth

New uses for mycelium are being discovered all the time! There's no limit on exploration with different substrates, which mushroom you use, and growth methods.



EXPERIMENTS IN FORMS (USING GROBIO STRAIN):

Substrate

My first experiment was surrounding substrate materials and growth forms. I looked around my kitchen, did some research online and sourced

I conducted 6 material experiments for different mycelium substrates, steaming them for two hours on the stove to sterilize.



1. Straw (pet bedding)

2. Coffee grounds

3. Oat pulp (oat-milk byproduct)

4. Shredded egg carton

5. Sawdust (bedding)

6. Oat pulp and sawdust blend

Substrate takeaways:

The coffee grounds molded right away, and eventually the oat pulp did too. The substrate needs move structure and airflow than those two materials allow; the egg cups, shavings and straw did much better and did not mold.





I wanted to design acoustic panel that would grow on, attach to and be reinforced by the mold itself. My first step was to build a thin wooden frame with a light cardboard backing.





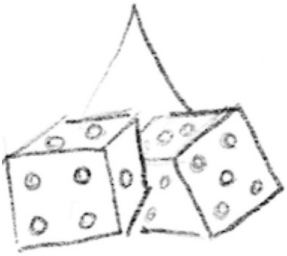
Then, I broke up the myclelium clumps as much as I could to fill in the mold. I created a humidity tent out of attached trash bags and added some books on top of the mycelium to create a flatter surface.

I left the mycelium growing in the mold for 5 days, adding more flour onto the surface on day 4. Ultimately, I left the mold covered for slightly too long; the mycelium started to brown in places.

I also put the mycelium against a heater and allowed it to dry too quickly. This caused the panel to warp slightly. The mycelium also shrinks considerably during the drying process, causing the wooden sides, which were attached to the mycelium, to bend inwards. This is not a problem for me. For furture versions, I would weigh down the edges of the panel to prevent warping.





I ran another experiment trying to make some deluxe fluffy mycelium dice for my car.

I used a silicon mold for XL square ice cubes. I realized through this process that in my previous experiments, I hadn't been packing my substrate into the mold tightly enough.



I also learned a lot about keeping a close eye on the mycelium to know when it's been growing for too long. After too many days, the mycelium starts to turn from powdery white to grey and brownish in patches.



I ran some experiments where I blended the innoculated hemp substrate so it would be finer when packed into the mold. The result was a much more precise edge on the corners and smoother sides.

The final obstacle in my mycodice is the side of the cube that is open to the air. Because of the flexible silicon mold, the cube becomes slightly wider at the top, and the material there is not smooth. For final versions, I would 3D print a fine mesh mold (to make it breathable) with four sides.





>>..... stool top ..... >>





